INSTITUT FÜR UMWELTWISSENSCHAFTEN LANDAU

UNIVERSITÄT KOBLENZ · LANDAU

MOD 1b: Environmental Modelling

MODELLING THE SPREAD AND CONTROL OF CHERRY GUAVA ON LORD HOWE ISLAND

PROJECT 3

David Nwoye

Course Lecturer: Lucas Streib

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1. Introduction

Invasive species are species that are not native to a specific location and can spread to degrees believed to cause damage to the environment, human economy, or health (Ehrenfeld, 2010). The fact that the invasive species are capable of outcompeting or consuming native species demands very effective ways of controlling them, especially in areas or islands harbouring special flora and fauna. Successfully controlling invasive species is a vital aspect of global conservation. Although there are challenges involved in invasive species eradication, modelling happens to be the major approach that is increasingly used to tackle these challenges. Examples of model approaches used include

- Optimizing search or control effort spatially
- Optimizing control effort temporarily
- Optimizing efforts across space and time simultaneously

1.1. Why are islands especially threatened by Invasive Species?

The Lord Howe Island, just like many other islands located in isolated areas are highly prone to invasive species due to the lack of natural competitors and predators that help in controlling population in their natural or native ecosystems. Secondly Islands usually have ecological niches that have not been occupied because of the distance from possible colonising populations, as a result increasing the probability of successful invasion (Reaser et. al., 2007)

1.2. The problem

Lord Howe Island is a world listed area located on the eastern coast of Australia. The Island consists of many unique species endemic to the island, which are also threatened species. The widest spread of species predicted to pose a more serious threat to the Island is the cherry guava plants. These plants were distributed on the Island by early travellers and were further distributed by birds and pigs. Although pigs are no longer spotted on the Island, the cherry guava seeds remain. Since the beginning of the weed

management program, 700,000 cherry guava plants from all life stages have been removed; however, more work is needed to entirely eradicate the weed population of cherry guava. Therefore, our research question is based on how much removal (per year) is needed to control the cherry guava crop.

1.3. Aims

Therefore, this project aims to provide decision support for cherry guava management on Lord Howe Island through the development of quantitative modelling. The key finding is the removal rate, which would be needed to eradicate the cherry guava plant.

2. Methods

For the modeling of the population development of the cherry guava plant, a matrix model was used. Matrix models are an elegant way for computing the population growth for species, in which age stages or life stages play an important role. A central aspect is the possibility of the computation of the finite growth rate (λ) and the stable stage distribution (SSD). Further, a stage-structured Lefkovitch matrix with three life stages was used: seedlings, juveniles, and matures (Lefkovitch, 1965; Pesendorfer et al., 2017). All parameters used in our model were provided by Baker et al. (2018). As shown in Figure 1, seedlings and juveniles can advance to the next stage or survive in their current stage for each yearly time step. Matures have reached the final growth stage. They can survive and produce seeds, which will lead to the recruitment of new seedlings. The life stage of seeds was left out, since the cherry guava plant doesn't produce seed banks (Baker et al., 2018). The values of the parameters, which can be seen in Figures 1 and 2, are: s1 <- 0.27, s2 <- 0.272, s3 <- 0.305, g1 <- 0.398, g2 <- 0.366, f3 <- 9.74.

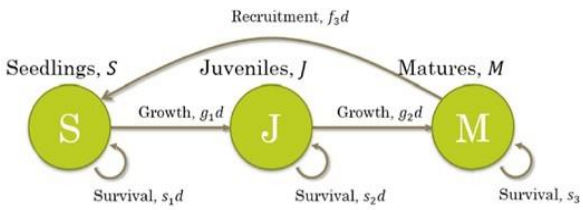

Figure 1. Conceptual model (Baker et al. 2018)

$$\begin{pmatrix} S_{t+1} \\ J_{t+1} \\ M_{t+1} \end{pmatrix} = \begin{pmatrix} S_t \\ J_t \\ M_t \end{pmatrix} * \begin{bmatrix} s_1 d & 0 & f_3 d \\ g_1 d & s_2 d & 0 \\ 0 & g_2 d & s_3 \end{bmatrix}$$

Figure 2. The equation with the Lefkovitch Matrix used in our model

The first step involved building a matrix model in R without density dependence (d=1) and without a removal term. The initial abundances were set to 1000, 100, and 10 for the seedlings, juveniles, and mature plants. In addition, we calculated λ and the SSD. This was done by calculating the first eigenvalue and the first eigenvector of the Lefkovitch matrix. Secondly, we included 10% and 35% removal. This was done by multiplying the equation of Figure 2 by the term “1-R”, where R stands for the removal rate expressed as a fraction (0.1 and 0.35). Lastly, an equation of the density-dependent parameter provided by Baker et al. (2018) with an area (A) of 9 hectares was added:

$$d_i = 1 - \frac{(\gamma_S S_i + \gamma_J J_i + \gamma_M M_i)}{A_i}$$

In each model step, we plotted the population development over a time of 25 years, and then observed how parameter variation affects the population dynamics. Furthermore, the result of λ was used to pinpoint the exact threshold of the removal rate, which decides about a positive or negative growth of the population. This was done by this equation:

$$\lambda(1-R) = 1$$

$$R = 1-(1/\lambda)$$

3. Results and Discussion

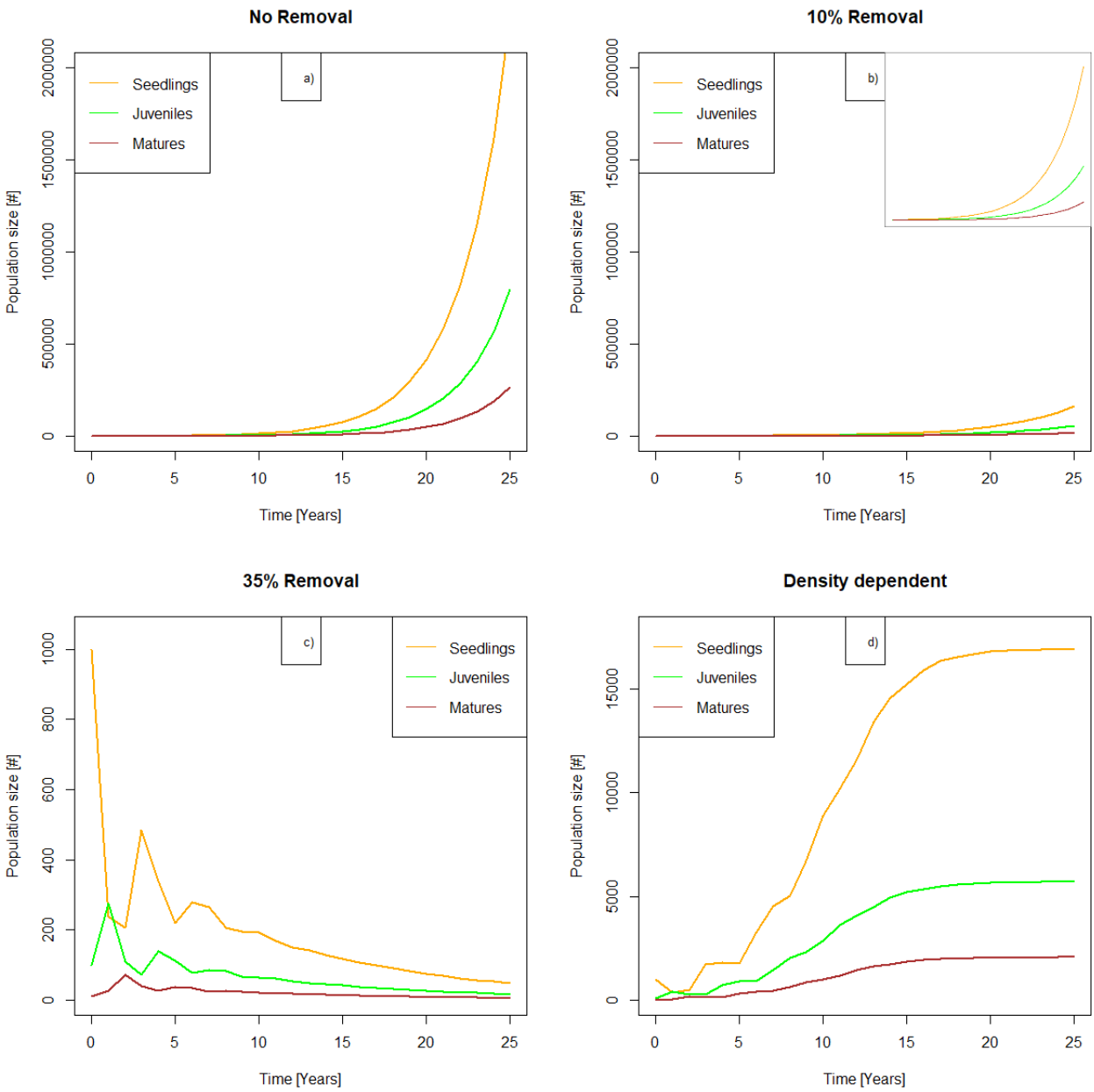

Figure 3. The population dynamics for cherry guava under different scenarios (a-d).

The first scenario (fig. 3a), without removal and density dependence, shows an exponential growth. The growth rate is so high that the population size of the seedlings grows from 1000 to over two million within 25 years. The value for λ is 1.4, and the values of the SSD are 0.68, 0.24, and 0.08. In the second scenario, the introduction of 10% removal results in a drastic decrease of growth (fig. 3b). However, it's also an exponential growth, which is more easily visible in the small window of figure 3b. At the end, the population size of the

seedlings are still over 160.000. In contrast, a removal rate of 35% is large enough to counter the exponential growth and results in an eradication of the population (fig. 3c). Adding the density parameter (and without the removal term) results in the plot of figure 3d. The former exponential growth turns into a logistic growth. The result of the threshold of the removal rate is 28.88%. A combination of this removal rate and initial abundances in accordance with the SSD leads to the equilibrium scenario in Figure 4.

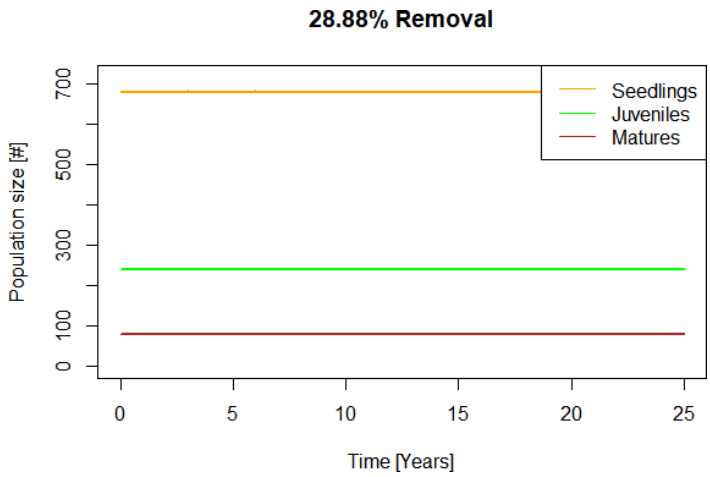

Figure 4. Equilibrium scenario. Initial abundances are 681, 239 and 79.

Some aspects are interesting enough to discuss further. Especially to get a deeper insight into the model. The fluctuations of the population sizes at the beginning, visible in Figures 3c and 3d, are due to the deviation of the population sizes from the SSD. Over time, the population sizes approach the SSD, and the fluctuations soften. These fluctuations are also present in Figures 3a and 3b. They are just not visible, since the scale of the y-axis is too large. Another interesting aspect is the density parameter (d) and its influence on the Lefkovitch matrix. With increasing population sizes, the parameter d decreases (see equation in Chapter 2). Since d is part of the Lefkovitch matrix, the matrix properties themselves change. Figure 5 shows how λ, the first eigenvalue of the matrix, is influenced by a decrease in d. The value of λ approaches 1, which means an approach to zero growth. This is one way of explaining the logistic growth of Figure 3d. These findings illustrate the significance of the eigenvalues and eigenvectors for the understanding of matrix models.

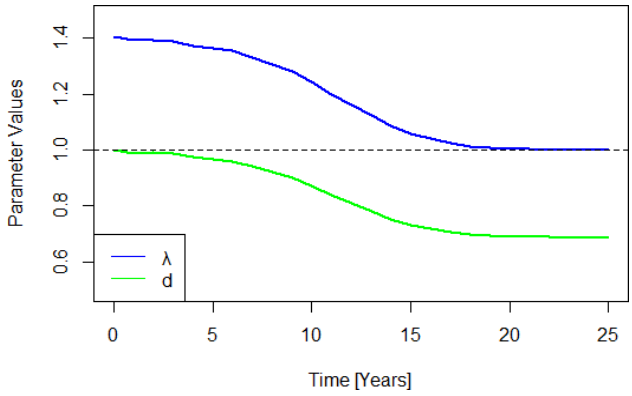

Figure 5. Influence of d on λ. Same scenario as in Figure 3d.

4. Conclusion

This model project looks at the control measures and how to prevent further spread of the cherry guava crop on Lord Howe Island. This is particularly important for forest managers, considering the spatial extent of this crop and the serious threat it poses to endemic species. Projects that are geared towards the management of invasive species are complicated and therefore, at each stage demands careful planning. Quantitatively, the knowledge and effect of various management strategies on project goals can be challenging. Hence, modeling can genuinely contribute to this vital aspect of conservation management (Baker et al., 2018). A significant contribution to this project is the finding of an appropriate removal rate (> 28.88%). The model output shows that such a removal rate will lead to an exponential decrease in the population size and further extinction of the plant population. Although it may not be realistic, some useful information from this model can assist in forest conservation management strategies. This point is supported by Box (1987), who stated that “essentially, all models are wrong, but some are useful”. In addition, models such as this have the challenge of limited available data. According to Grantham et al. (2009), a trade-off exists between the collection of information and the delay of management decisions or swift decision-making with less information for all conservation projects. However, proper care must be taken because species can have a poor outcome if intervention is delayed for too long (Lindenmayer et al., 2013).

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